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The Pitfalls of Review Solicitation: Evidence from a Natural Experiment on TripAdvisor

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Abstract. This study examines the effect of firms' participation in platform-endorsed review solicitation programs on consumers' online review generation. We leverage a natural experiment on TripAdvisor, which launched a review solicitation program that allows hotels to collect reviews directly from guests after their stays with the aid of certified connectivity partners. Applying a two-stage difference-in-differences approach to a panel data set of online reviews for a matched set of hotels across TripAdvisor and Expedia, we find that hotels' participation in the review solicitation program results in a 34.3% increase in review volume, a 0.151 increase in review rating, but a 16.9% decrease in review length. Review solicitation, however, generates a notable negative spillover effect on the volume of organic reviews. Specifically, the volume of organic reviews is reduced by 15.5% after hotels start soliciting reviews. We provide evidence that the motivational crowding-out effect plays an important role in driving this negative spillover. Further analyses reveal that the effects of review solicitation are heterogeneous with respect to hotels of different types and consumers with different demographic and behavioral characteristics. Finally, using a novel structural topic model, we detect a significant shift in review content from specific and concrete topics to general and abstract topics. Our findings suggest that review platforms and firms should be cautious about the unintended negative consequences of review solicitation on consumers' review generation.

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Keywords: review solicitation program • online reviews • two-stage difference-in-differences • spillover effect • structural topic model

1. Introduction

Consumers today rely heavily on online peer reviews to guide their purchase decisions. It was reported that 65% of consumers trust online reviews more than the information provided by advertisers (MacKinnon 2012), and 79% of consumers trust online reviews as much as personal recommendations from people they know (Murphy 2020). For review platforms, user-generated reviews are at the core of their business models and are thus critical for their value and sustainability. For businesses, online reviews are an important source of consumer voice and have been shown to be a key driver of product and service sales (Chevalier and Mayzlin 2006, Duan et al. 2008, Zhu and Zhang 2010, Rosario et al. 2016).

Yet, consumers' underprovision of online reviews is a major challenge faced by many review platforms and businesses. More concerningly, there has been a steady decline in the average length of reviews over time from 600 characters in 2010 to 200 characters in 2017 (ReviewTrackers 2022). To address these problems, many businesses have begun to incentivize consumers to write reviews (or even positive reviews) in exchange for economic rewards such as coupons, discounts, gifts, free products, and direct payments without making any disclosure (Murphy 2020). Although such practices can be very effective in boosting review volume, they are costly and against both the regulations of the Federal Trade Commission and the guidelines of major review platforms such as Google, Yelp, and TripAdvisor. In addition, offering financial incentives may lead to low-quality reviews and irritate other consumers when such incentives are identified (Stephen et al. 2012, Burtch et al. 2018, Magno et al. 2018).

Many review platforms are also fighting against the dearth of consumer reviews by adopting a variety of approaches. For example, Amazon launched its Vine program to help participating vendors send free products to a select group of trusted customers in exchange for their reviews.[1] TripAdvisor implemented a Review Express program, which facilitates the review collection of participating hotels by sending emails to guests automatically after their stays with the aid of certified connectivity partners.[2] Compared with business-initiated review collection, the approaches adopted by review platforms are more legitimate because of their strict policies against review fraud and explicit disclosure of solicitation information. However, platforms such as Yelp oppose any form of review solicitation, including offering incentives or freebies or simply sending review solicitation emails, claiming that such practices would lead to biased reviews and mislead consumers.[3] This poses an interesting question: Is it beneficial for businesses to participate in such platform-endorsed review solicitation programs?

The answer is not straightforward. First, although some suggestive evidence has shown that participation in platform-endorsed review solicitation programs leads to more consumer reviews, we lack rigorous empirical investigations to causally assess the effect of such solicitation on review generation. Furthermore, even if the participation is effective in inducing consumers to contribute *solicited reviews* and results in an increase in the total review volume, its spillover effect on the volume of *organic reviews* that are written by consumers without any solicitation is not yet understood. On the one hand, more reviews can enhance firms' popularity and visibility as well as attract more consumers, who can then provide more organic reviews. On the other hand, review solicitation and the presence of solicited reviews might reduce the number of organic reviews due to either a substitution or a motivational crowding-out effect.

Second, solicitation might change the characteristics of consumer reviews, such as valence, length, and topics. The characteristics of solicited reviews can be substantially different from those of organic reviews due to motivational changes of consumers when being asked to review. Furthermore, the presence of solicited reviews might affect the characteristics of subsequent organic reviews as a result of social influence (Muchnik et al. 2013) or attitude change. Such changes in review characteristics can affect the value of the reviews in assisting consumers' decision making.

Third, the overall and spillover effects of review solicitation may vary with different types of businesses. The review solicitation of superior businesses (e.g., higher popularity, better quality, and larger size) may elicit different consumer responses from that of inferior ones, resulting in considerable heterogeneity in treatment effects. Similarly, consumers with different demographic and behavioral characteristics may be heterogeneous in their propensity to write solicited reviews and deviation in review characteristics on participation. The lack of theoretical and empirical investigations suggests a gap in our knowledge.

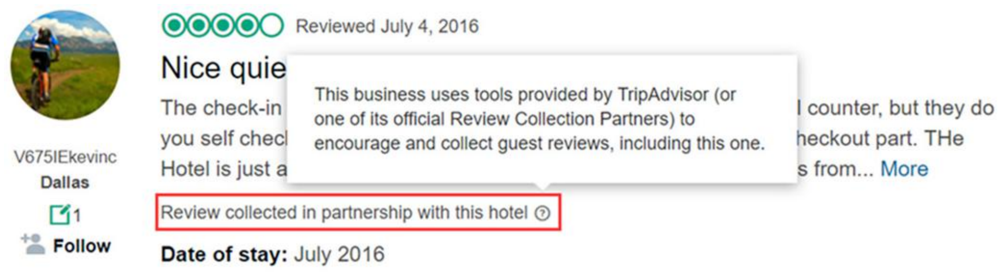
This study aims to shed some light on this important yet underexplored question in the above aspects by taking advantage of TripAdvisor's launch of a review collection program in 2012, which enables hotels to send customized emails to collect reviews from their guests using tools provided by TripAdvisor or its review collection partners and seamlessly publish the reviews on its website with explicit solicitation disclosure (for an example of a solicited review on TripAdvisor, see Figure 1). We empirically investigate the effects of hotels' review solicitation on consumers' generation of hotel reviews and explore the underlying mechanisms.

Given that hotels self-select into the treatment group and use customized emails in their review solicitation, we can only estimate the average treatment effect on the treated (ATT). The ATT will be biased if a hotel's decision to participate is driven by unobserved factors that also affect its reviews, such as interior/exterior renovations and service quality improvements. To address the potential endogeneity caused by unobserved hotel actions, we match hotels on TripAdvisor with those on Expedia, another popular travel platform, and use a two-stage difference-in-differences (DID) approach to compare reviews of the same hotels on the two platforms before and after the hotels received the treatment (i.e., participated in TripAdvisor's review solicitation program). We assess the parallel trends assumption empirically to ensure the validity of this DID specification.

Our cross-platform two-stage DID estimations yield several important findings. First, participating in the review solicitation program led to a 34.3% increase in the number of total reviews. However, it is worth noting that this comes at the cost of a 15.5% reduction in the number of organic reviews. We provide evidence that the negative spillover of review solicitation to consumers' contribution of organic reviews cannot be explained by an intuitive substitution effect. Instead, the motivational crowding-out of organic reviewers (i.e., reviewers whose first reviews were organic) plays an important role in driving this negative spillover.

Second, on average, review valence increased by 0.151 in star rating, and review length decreased by 16.9% in number of characters after hotels participated in the solicitation program, which indicates that solicitation hurts the value of consumer reviews, in light of previous findings that people tend to perceive shorter

Figure 1. (Color online) Sample Solicited Review on TripAdvisor



reviews and positive reviews as less useful (Mudambi and Schuff 2010, Chen and Lurie 2013, Casaló et al. 2015). Review solicitation did not appear to result in significant changes in the valence and length of subsequent organic reviews.

Third, we reveal the heterogeneous effects of review solicitation across different types of hotels. Despite the fact that chain, larger, more expensive, higher-class, more popular, and higher-rated hotels are more likely to participate in the solicitation program, our results show that they benefit less from the participation in terms of the lift in review valence compared with independent, smaller, cheaper, lower-class, less popular, and lower-rated hotels. We do not observe any significant heterogeneous effects across different types of hotels with respect to review volume and review length. Furthermore, we do not find any significant heterogeneity in the spillover effect.

Next, we conduct additional analyses at the reviewer level to investigate the potential heterogeneities in their propensity to write solicited reviews and deviation in review characteristics from organic reviews. We find that reviewers who are female, older, nicer, less reputable, less privacy conscious, and come from individualistic cultures are more likely to write solicited reviews. Additionally, the reviews written by females, older individuals, and lower-level reviewers, as well as those from individualistic cultures, are influenced more by the solicitation in rating inflation and/or length reduction.

Last, we use a novel structural topic model (STM) to mine the textual content of solicited and organic reviews and assess the effect of review solicitation on the topic distributions of reviews. The results indicate that solicitation results in a significant shift in review content from specific and concrete topics to general and abstract topics. Specifically, reviewers are likely to comment on specific details of hotels such as services, facilities, design, and food when writing organic reviews. However, they tend to focus on their attitudes toward hotels when being solicited to contribute reviews. The result is corroborated by the Generative Pre-trained Transformer (GPT) model and remains significant even after excluding the mediation effect of rating and length. This shift in review topics is likely to hurt the depth and perceived diagnosticity of consumer reviews (Mudambi and Schuff 2010).

This paper contributes to research on online reviews in several important ways. First, we expand the literature on the effect of different solicitation strategies on consumers' review generation. Despite the abundance of studies on the role of financial incentives (e.g., free products, rebates, bonus points, gifts, direct payments) on recipients' review contributions at the seller level (Cabral and Li 2015, Wang et al. 2016, Burtch et al. 2018) and at the platform level (Khern-am-nuai et al. 2018, Petrescu et al. 2018, Qiao et al. 2020, Yu et al. 2022, Pu et al. 2023), there is a dearth of research that focuses on the impact of platform-endorsed nonincentivized review solicitation with explicit disclosure. By leveraging a policy shock on a leading review platform and applying a cross-platform two-stage DID identification approach, we are able to rigorously quantify the effect of such a solicitation strategy on consumer reviews. Our findings—that the solicitation results in more reviews of inflated ratings and shorter lengths—add new empirical evidence to the literature in this area.

Second, we provide novel insights into consumers' responses to review solicitation by examining the indirect spillover effect of review solicitation on organic reviews. Prior work on the spillover effect of review solicitation has mainly focused on how receiving financial incentives for writing reviews influences receivers' subsequent organic reviews (Qiao et al. 2020, Yu et al. 2022, Pu et al. 2023). In this work, we complement this line of research by exploring the spillover effect of review solicitation on all organic reviews, regardless of whether they are from reviewers who have or have not written solicited reviews. Although review solicitation does not significantly affect the valence and length of organic reviews, it generates a strong negative spillover effect on the volume of organic reviews. We provide supporting evidence that the motivational crowding-out effect plays an important role in driving this negative spillover. The results deepen our understanding of consumers' motivational and behavioral responses in their review generation to review solicitation.

Third, existing research on review solicitation has examined the effects at the aggregate level without attending to the contextual nuances. Little is known as to how the effects may vary across different types of businesses. We advance a novel contribution to the literature by exploring this heterogeneity at the business level. We find that chain, larger, more expensive, higher-class, more popular, and higher-rated hotels benefit less from the review solicitation, which adds to the existing knowledge of the use of review solicitation by businesses. There is also no or little knowledge on how individuals with various characteristics might respond to review solicitation. We reveal that consumers who are female, older, nicer, less reputable, less privacy-conscious, and originate from individualistic cultures, exhibit a higher propensity to write solicited reviews. Furthermore, the solicited reviews composed by female, older, and lower-level reviewers, as well as those from individualistic cultures, deviate more from their organic reviews in rating and/or length. These nuanced findings provide valuable insights into the literature on the effects of review solicitation.

Last, we go beyond the previous research on review solicitation by mining the textual content to identify the topics covered by consumer reviews. Past work in this area has mainly focused on review volume, valence, length, helpfulness, and sentiment (Burtch et al. 2018, Khernam-nuai et al. 2018, Qiao et al. 2020, Yu et al. 2022). By using a novel STM model, we detect a significant shift in review content from specific descriptions of one's experience to general unsubstantiated comments. This finding is alarming in that both the informativeness and the value of reviews would deteriorate substantially, which could have long-term and irreversible consequences for the sustainability of the platform and the attractiveness of businesses. By tapping into a new and important dimension of review changes, we discover a pitfall of review solicitation that is usually overlooked in the literature.

The rest of this paper is organized as follows. Section 2 reviews the previous literature. Section 3 describes the research context and the data. Section 4 explains the main identification strategies and empirical specifications. Section 5 presents the results and conducts the robustness checks. Section 6 explores the underlying mechanisms. Section 7 conducts additional analyses. Section 8 concludes and discusses the implications and limitations of our work.

2. Related Literature

During the past two decades, a vast amount of research has been devoted to understanding why consumers are willing to spend their time and effort writing online reviews. One major driver of the review contribution is altruism, where consumers share their consumption experiences and opinions with the desire to help others with their purchase decisions (Sundaram et al. 1998, Hennig-Thurau et al. 2004). In addition to altruism, image motivations such as peer recognition or reputational gains in the community may also motivate consumers to write reviews (Goes et al. 2014, Shen et al. 2015, Huang et al. 2017, Wang et al. 2019). Previous studies have also cited social norms or influence as a significant factor in motivating online review contributions (Chen et al. 2010, Burtch et al. 2018, Ke et al. 2020). Recently, more research has begun to examine the role of consumers' economic incentives, showing that consumers are extrinsically motivated by different forms of economic rewards such as direct payments (Wang et al. 2012, Burtch et al. 2018), loyalty points (Wang et al. 2016, Khern-am-nuai et al. 2018), rebates (Cabral and Li 2015), free products (Kim et al. 2019, Qiao et al. 2020, Pu et al. 2023), and gifts (Yu et al. 2022).

Our research is closely related to several recent studies devoted to investigating the effects of review solicitation strategies on various aspects of consumer-generated online reviews, such as volume, valence, length, and helpfulness. The vast majority of papers in this stream of literature have been dedicated to the direct effect of review solicitations with economic rewards. Although the evidence for the positive impact of monetary incentives on review volume is consistent and pervasive, the findings regarding their effects on review valence, length, and helpfulness are quite mixed. For example, researchers have shown that the use of monetary incentives may variously have a positive effect (free products: Pu et al. 2023; performance-contingent payment: Wang et al. 2012), negative effect (loyalty points: Khern-am-nuai et al. 2018), or no effect (direct payment: Burtch et al. 2018) on the length of stimulated reviews. The ratings of these rewarded reviews could be higher than (free products: Mo and Li 2018, Lin et al. 2019; rebates: Cabral and Li 2015; loyalty points: Khern-am-nuai et al. 2018), lower than (free products: Pu et al. 2023), or similar to (loyalty points: Wang et al. 2016; free products: Petrescu et al. 2018) reviews written organically. In addition, the rewarded reviews may be perceived as more helpful (performance-contingent payment: Wang et al. 2012), less helpful (free products: Pu et al. 2023; loyalty points: Khern-am-nuai et al. 2018), or similarly helpful (loyalty points: Wang et al. 2016) compared with organic reviews.

Another set of papers focuses on the effect of review solicitation via email invitations without any financial incentives. Using a questionnaire-based survey, Magno et al. (2018) find that although soliciting reviews may induce more reviews for businesses, it also irritates a significant share of guests, especially when a business explicitly asks for positive reviews. Karaman (2020) finds that solicitations can significantly increase the number of reviews and improve the representativeness of rating distributions. Litvin and Sobel (2019) make a simple comparison between a random set of organic and solicited reviews and report that solicited reviews are shorter and have higher ratings. Similarly, Askalidis et al. (2017) report that solicited reviews are more positive, shorter, and less helpful. They also demonstrate that email invitations can reduce both the social influence bias and the selection bias present in online review systems. Our paper adds to this body of literature by rigorously quantifying the overall impact of a platform-endorsed nonincentivized review solicitation program with explicit disclosure on consumers' generation of online reviews, which represents a unique setting that deserves closer attention.

Recently, several studies have begun to explore the indirect spillover effect of economic rewards on receivers' subsequent organic reviews. For example, Pu et al. (2023) find that reviewers generate more organic reviews that are more positive, longer, and more helpful after they receive free products. By contrast, Qiao et al. (2020) show that reviewers who have received free products in the past tend to write fewer and shorter organic reviews, with an upward sentiment bias. In a different setting, Yu et al. (2022) find that the receipt of performance-contingent rewards leads to a significant increase in both the volume and quality of customers' subsequent organic reviews. The rating of these reviews, however, is not significantly affected. The influence of review solicitation, however, may extend to consumers who do not contribute solicited reviews as they may be aware of the solicitation either through receiving solicitation emails or seeing solicited reviews with explicit disclosure. Mo and Li (2018) pursue this direction and find that free product sampling leads to longer reviews with lower ratings from regular reviewers. We extend this line of research by investigating the spillover effect of review solicitation on the volume and characteristics of all organic reviews in a more legitimate setting where financial incentives are prohibited and solicited reviews are clearly labeled with a disclaimer.

In addition to rigorously quantifying the overall and spillover effects of review solicitation, we aim to advance the literature in several other aspects. First, prior work on review solicitation (Khern-am-nuai et al. 2018, Qiao et al. 2020) thus far has been looking at the aggregate effect without considering the heterogeneity of businesses. We address this issue by exploring how these effects of review solicitation may vary across different types of businesses. Second, existing literature (Burtch et al. 2018, Karaman 2020) has treated consumers as a whole and ignored their demographic and behavioral dissimilarities. We make the first empirical attempt in this area to understand the potential heterogeneities across reviewers in the propensity to write solicited reviews and deviation in review characteristics upon participation. Third, past studies investigating the role of solicitation in online review generation have predominately focused on three aspects: volume, valence, and quality (proxied by length and helpfulness). Recently, increasing research efforts have been directed to analyze the textural characteristics of reviews such as sentiment (Qiao et al. 2020, Yu et al. 2022), affect (Litvin and Sobel 2019), readability (Khernam-nuai et al. 2018), and lexical richness (Qiao et al. 2020). We go beyond these metrics by employing a novel topic modeling technique to assess whether solicitation causes any significant shift in the topic distribution of reviews. This allows us to offer a fresh perspective on the effects of review solicitation on consumer reviews.

3. Research Setting and Data

3.1. Study Context

Our research setting is TripAdvisor, the world's largest travel website. As of March 2023, TripAdvisor had accumulated more than 1 billion reviews and opinions of nearly 8 million businesses.[4] The enormous volume of user-generated reviews not only helps millions of travelers each month across the globe plan and book their trips, but also allows businesses to attract more relevant customers and generate higher revenue. Despite the substantial benefits of online reviews, travelers' underprovision of reviews after their visit or travel poses a significant challenge to the value and sustainability of both the platform and businesses that fall short of consumer feedback. To help businesses collect more guest reviews, TripAdvisor launched a review collection program in 2012, which allows businesses to create and send customized review request emails to their guests themselves or with the help of certified connectivity providers. The solicited reviews are displayed on hotels' review pages on TripAdvisor with a disclosure clause stating, "Review collected in partnership with this hotel" or "Review collected in partnership with (the brand name of the hotel)" at the end of reviews. Such a disclaimer enables consumers to distinguish solicited reviews from organic reviews easily.

3.2. Data

Our data sample consists of hotels located in the four largest cities in Texas (Houston, San Antonio, Dallas, and Austin) from two major sources (TripAdvisor and Expedia). We first collected all the guest reviews posted for 1,138 hotels listed on TripAdvisor up to January 2017, which covers a total of 348,653 reviews, with the earliest one dating back to September 2001. For each review, we recorded the reviewer ID, submission date, star rating, text content, and disclaimer that indicates whether this review was collected in partnership with the hotel (if any). Of the 1,138 hotels in this sample, 554 hotels had received at least one solicited review, which is used as the proxy for the hotel's participation in the review solicitation program. We treat the earliest date that a hotel received a solicited review as the hotel's participation date.

Our second data source is Expedia, another popular travel platform that hosts a huge number of user-generated reviews. Similarly, we collected all guest reviews posted for 1,780 hotels that are listed on Expedia from September 2004 to January 2017, for a total of 861,493 reviews. We recorded the same information for each Expedia review as for the TripAdvisor data, with the exception of the disclaimer, as Expedia does not have a review solicitation program.

Next, we matched the hotels on TripAdvisor with those on Expedia by utilizing the link provided by TripAdvisor to the hotel's Expedia page (if such a page exists). We randomly selected 100 TripAdvisor–Expedia pairs and verified the link by checking the names and addresses of matched hotels. We found them all accurate. Following this procedure, we were able to obtain a total of 535 matched hotels that participated in TripAdvisor's review solicitation program. Of the 535 matched hotels, 12 hotels had no review on Expedia until one year after their participation in the review solicitation program. After excluding these 12 hotels, we are left with 523 matched hotels and 748,060 reviews, of which 279,520 are from TripAdvisor and 468,540 are from Expedia.

Finally, we aggregate the review data to the hotel-platform-month level and focus our analyses on the 12 months before and after hotels' review solicitation on TripAdvisor, which gives us a total of 23,350 observations.[5]

3.2.1. Descriptive Statistics.

Hotels differ in their decisions and time to take part in TripAdvisor's review solicitation program. Figure 2 presents the number of hotels in our sample that started to participate in the program over time. As we can see, the majority of participating hotels joined the program in 2012 or 2013. Table 1 provides a comparison between participating and nonparticipating hotels on TripAdvisor in terms of various hotel characteristics. Overall, chain, larger, more expensive, higher-class, more popular, and higher-rated hotels are more likely to participate in the review solicitation program. Figure 3 shows the proportion of solicited reviews for the 554 participating hotels on TripAdvisor over time. It is apparent that the share of solicited reviews increased substantially in our observation period. Table 2 presents a comparison between organic reviews and solicited reviews in terms of average star rating, average number of characters, and average number of helpful votes. Compared with organic reviews, solicited reviews are more positive, are shorter, and receive fewer helpful votes, which are in line with the preliminary findings of Litvin and Sobel (2019) that solicited reviews are more positive and shorter than organic reviews.

3.2.2. User Review History.

In later sections, we use the entire review history of reviewers on TripAdvisor for additional analyses. For each user who reviewed a hotel

Figure 2. Number of Hotels That Started to Participate in Review Solicitation over Time

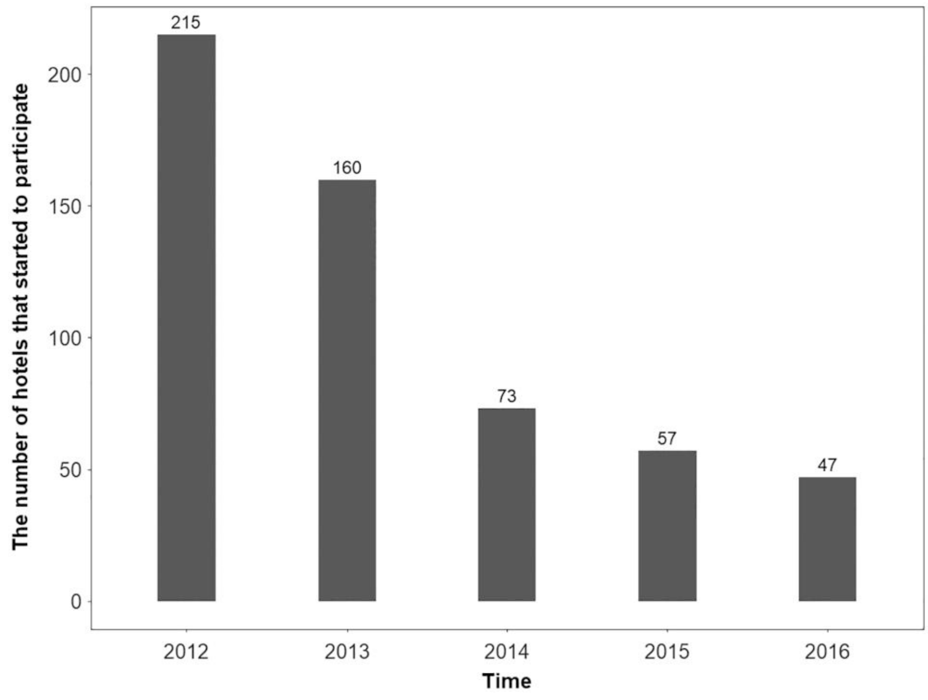


Table 1. Participating Hotels vs. Nonparticipating Hotels

Variable	Participating hotels	Nonparticipating hotels
Ratio of chain hotels	0.80	0.68
Average hotel size (in number of rooms)	166.87	114.28
Average price per night ($)	156.01	120.99
Average star class	2.87	2.56
Average review volume (in the first half year)	12.81	5.58
Average review rating (in the first half year)	3.58	3.21

in our sample on TripAdvisor, we collected all the hotel reviews that this reviewer submitted in the past, including reviews written for hotels outside our sample. This gives us a total of 2,588,481 hotel reviews from 268,092 reviewers.

4. Empirical Strategy

Our objective is to estimate the ATT, which focuses on the effects of review solicitation on the reviews of hotels that participate in the review solicitation program. This is different from the average treatment effect (ATE), which measures the effects of review solicitation on the reviews of a randomly selected set of hotels. The estimation of ATE is difficult, as we cannot force a random set of hotels to participate in TripAdvisor's review solicitation program. Furthermore, there might be considerable variation in hotels' implementation of the review solicitation (e.g., the messages they use in the solicitation emails may vary significantly). The ATT that we estimate in this paper measures the effects of review solicitation conditional on the hotels that choose to participate in this program and the customized emails they actually use in their solicitation.

4.1. Cross-Platform Two-Stage DID Model

To accurately assess the ATT, ideally we need the counterfactual reviews that consumers would have contributed had they not been exposed to review solicitation, which is impossible. In this paper, we leverage the reviews of the same hotels on Expedia as the control group. Intuitively, review solicitation should affect the reviews of participating hotels on TripAdvisor but not those of the same hotels on Expedia.[6] We use a cross-platform DID identification strategy, which compares changes in a hotel's TripAdvisor reviews following its decision to solicit reviews with changes in the same hotel's Expedia reviews over the same time period. This strategy allows us to control for the unobserved shocks to hotel reviews (e.g., improvements in service quality) that are common to both platforms. The main identification assumption is, in the absence of review solicitation, differences between TripAdvisor and Expedia reviews would have been constant over time, which is often referred to as the "parallel trend assumption." In Section

Figure 3. Proportion of Solicited Reviews on TripAdvisor over Time (Only 554 Participating Hotels)

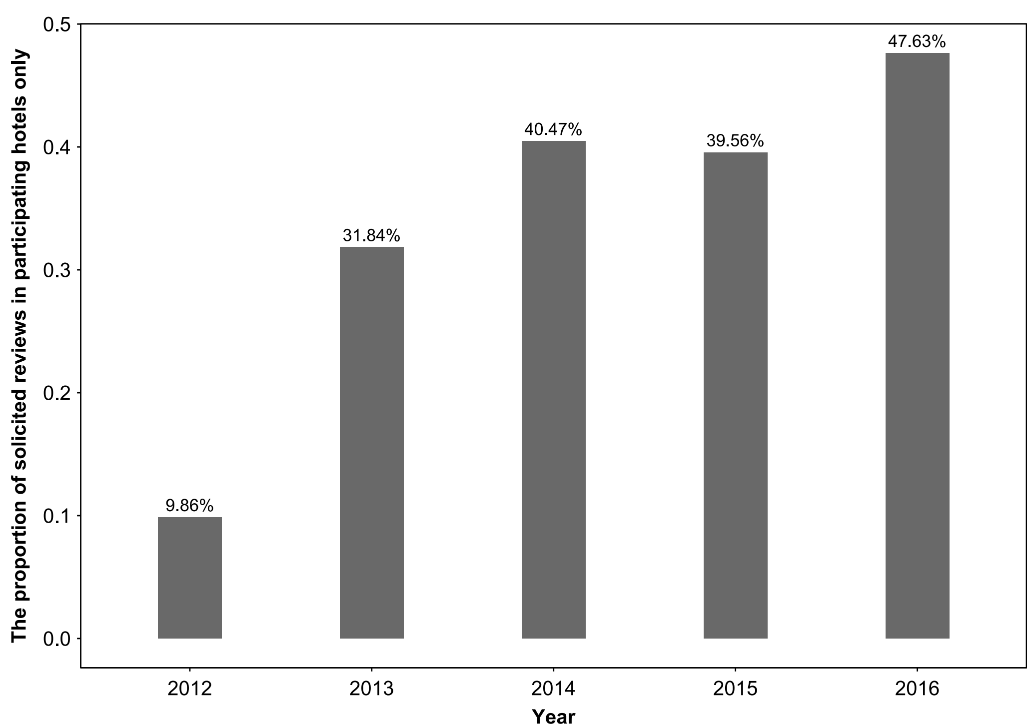


Table 2. Organic Reviews vs. Solicited Reviews

Test variable	Statistics	Organic reviews	Solicited reviews	t	p
Star rating	Mean	4.03	4.07	−7.43	0.000
	Standard deviation	1.16	1.21		
Number of characters of review	Mean	579.51	286.96	218.18	0.000
	Standard deviation	501.25	202.01		
Number of helpful votes of review	Mean	0.57	0.19	110.09	0.000
	Standard deviation	1.31	0.50		

Note. The average number of characters includes only English language reviews.

5.3.1, we use a relative time model (specified in Section 4.2) to validate this assumption by comparing the pretreatment trends of reviews on TripAdvisor and Expedia with respect to various outcomes of interest. We use the following model specification to implement our cross-platform DID identification strategy:

$$DV_{ipt} = \beta_1 After_{it} + \beta_2 TripAdvisor_p + \delta After_{it} \times TripAdvisor_p + \alpha_i + \tau_t + \varepsilon_{ipt}, \quad (1)$$

where DV_{ipt} variously denotes the log number of reviews, average rating of reviews, and log average length of reviews for hotel i in month t on platform p. $After_{it}$ is a dummy variable that equals one if hotel i has started review solicitation on TripAdvisor in or before month t and zero otherwise. $TripAdvisor_p$ is an indicator for reviews on TripAdvisor. We include a matched-pair fixed effect α_i for a shared fixed effect for the same hotel i on either platform. We also include year-month fixed effects τ_t to control for temporal shocks that are common to all hotels in a month t. Finally, ε_{ipt} captures any idiosyncratic random errors. With this specification, the coefficient δ is the DID estimate of interest, which captures the effect of review solicitation on review volume, rating, and length.

Given that hotels' participation in the review solicitation program is staggered and the treatment effects could vary across hotels and over time, the conventional two-way fixed-effects (TWFE) model above may fail to identify the ATT. To address this issue, we use the two-stage DID estimator developed by Gardner (2021) to measure the effect. In this model, the first stage estimates the group and period fixed effects using the subsample of untreated observations, and the second stage estimates the ATT by subtracting the estimated group and period effects from observed outcomes and regressing the adjusted outcomes on treatment status. The two-stage DID estimator produces robust estimates of the ATT in the presence of staggered adoption and treatment effect heterogeneity.

4.2. Relative Time Model

To check whether consumer reviews on TripAdvisor and Expedia have parallel trends before review solicitation, we rely on a relative time model (Angrist and Pischke 2009) that replaces $After_{it}$ with a vector of relative time dummies that indicate the distance between month t and the month when hotel i started review solicitation. The model specification is as follows:

$$\mathrm{DV}_{ipt} = \sum_{m=-12}^{11} \gamma_m TD_{it,m} + \beta_2 TripAdvisor_p + \sum_{m=-12}^{11} \lambda_m TD_{it,m} \times TripAdvisor_p + \alpha_i + \tau_t + \varepsilon_{ipt}, \quad (2)$$

where $TD_{it,m}$ is a dummy variable indicating that the current month t is m months before (in the case of a negative m) or after (in the case of a positive m) the month when hotel i started soliciting reviews. The coefficients λ_m ($m = -12, \ldots, 12$) are our estimates of interest, which allows us to assess whether the parallel trend assumption holds before the review solicitation and how the effects of review solicitation manifest over time afterward. We extend the two-stage procedure described above by replacing the treatment status with a set of relative time dummies to estimate these coefficients.

4.3. Cross-Platform Difference-in-Difference-in-Differences (DDD) Model

As an alternative identification, we use a more stringent cross-platform DDD model using the hotels that have not participated in the review solicitation program as a control group of participating hotels. In the DDD specification, we compare changes in TripAdvisor reviews of participating hotels after review solicitation with changes in Expedia reviews of matched hotels (i.e., the DID estimate) and adjust this for unobserved platform-specific trends by subtracting out cross-platform changes in the reviews of nonparticipating hotels, all over the same period of time. The following specification implements our DDD identification:

$$DV_{ipt} = \beta_1 After_{it} \times Participating_i + \beta_2 TripAdvisor_p + \beta_2 TripAdvisor_p \times Participating_i + \delta After_{it} \times TripAdvisor_p \times Participating_i + \alpha_i + \tau_t + \varepsilon_{ipt}, \quad (3)$$

where $Participating_i$ is an indicator of hotels' participation in TripAdvisor's review solicitation program. The coefficient δ is our DDD estimator, which captures the effect of review solicitation on the volume and characteristics of reviews of participating hotels. Again, we follow the same two-stage procedure as described previously to produce robust estimate of this coefficient under staggered adoption.

5. Results

5.1. Effects on All Reviews

We first estimate the effects of review solicitation on all reviews posted for hotels using the model in Equation (1). To account for possible dependence among observations within hotels, we cluster standard errors at the hotel level. Column 1 of Table 3 presents the results for review volume. The estimated coefficient of $After_{it} \times TripAdvisor_i$ is positive and statistically significant. Specifically, hotels' participation in TripAdvisor's review solicitation program leads to a 34.3% increase in the number of reviews per month.[7] This is consistent with TripAdvisor's claim that "on average, regular Review Express users see an uplift of 28% in the amount of TripAdvisor reviews for their property" (TripAdvisor 2017). Columns 2 and 3 of Table 3 present the results for the rating and length of reviews, respectively. Review solicitation increases the average rating of reviews by 0.151 and decreases the average length of reviews by 16.9%. The set of results is not surprising, given that review solicitation brings in a substantial number of solicited reviews that are more positive and less detailed, causing significant changes in the overall distribution of review characteristics.

5.2. Spillover Effects on Organic Reviews

The previous results confirm that participation in the review solicitation program can be an effective way to boost the total volume of reviews. However, it is not clear how the practice of review solicitation, and the presence of solicited reviews that are vastly different from organic reviews and labeled with an explicit disclosure statement, would affect the volume and characteristics of subsequent organic reviews contributed by reviewers. Here, we explore the spillover effect of review solicitation on organic reviews posted on TripAdvisor.

We again estimate the cross-platform two-stage DID model but restrict our analysis to organic reviews only. The results are reported in Table 4. We find that the coefficient of the interaction term in Column 1 is negative and statistically significant, indicating that review solicitation causes a negative spillover effect on the volume of organic reviews written for hotels. Specifically, the volume of organic reviews is reduced by 15.5% after hotels start soliciting reviews. From columns 2 and 3, we see that review solicitation does not significantly affect the rating and length of organic reviews, which suggests that the effects on review valence and length in Table 3 are driven mainly by the influx of more positive and shorter solicited reviews rather than by changes in the valence and length of organic reviews.

5.3. Robustness Checks

5.3.1. Parallel Trend Assumption. We first use our relative time model in Section 4.2 to assess whether the parallel trends assumption underlying our identification strategy holds. We set the one-month interval right before the solicitation as the baseline to make the presentation clearer. Then, the coefficients of the remaining interaction terms capture the differences between TripAdvisor and Expedia reviews over time with respect to the month before the treatment. Figures A.2 and A.3 in the Online Appendix present visualizations of the coefficient estimates λ_m and their 95% confidence intervals for all reviews and organic reviews respectively. Two patterns in these dynamic trends add support to the validity of our results. First, the coefficient estimates associated with almost all the time interval interactions before zero are not significantly different from zero, in

Table 3. Effect of Review Solicitation on All Reviews (Two-Stage DID)

	Dependent variable		
	log(Volume) (1)	*Avg_Rating* (2)	*log(Avg_length)* (3)
After × *TripAdvisor*	0.295*** (0.029)	0.151*** (0.022)	−0.185*** (0.025)
TripAdvisor	Yes	Yes	Yes
After	Yes	Yes	Yes
Hotel fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Observations	23,350	20,122	20,000

Notes. Clustered robust standard errors at the hotel level are in parentheses. The average length includes only English language reviews.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

Table 4. Effect of Review Solicitation on Organic Reviews (Two-Stage DID)

	Dependent variable		
	log(Volume) (1)	*Avg_Rating* (2)	*log(Avg_length)* (3)
After × *TripAdvisor*	−0.169*** (0.026)	0.040 (0.022)	0.005 (0.023)
TripAdvisor	Yes	Yes	Yes
After	Yes	Yes	Yes
Hotel fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Observations	23,350	19,456	19,310

Notes. Clustered robust standard errors at the hotel level are in parentheses. The average length includes only English language reviews.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

support of the notion that the volume and characteristics of reviews on TripAdvisor move in parallel with those on Expedia prior to the treatment of review solicitation. This confirms that the same hotels on Expedia can serve as a valid control group. Second, the treatment effects manifest immediately after the review solicitation, suggesting that the identified effects are true and unlikely to be driven by spurious correlations.

5.3.2. Alternative Samples. Next, we assess the robustness of our findings with respect to the use of alternative samples. First, in our main sample, there are 114 hotels that do not have the full observations of 24 months on the two platforms, contributing to approximately 2.4% missing observations. The presence of missing values may bias our estimates. As a robustness check, we remove these hotels with missing observations and reestimate our models. Tables A.1 and A.2 in the Online Appendix present the estimation results. Once again, the results are qualitatively similar to those reported in Tables 3 and 4.

Second, the solicited reviews in some hotels may be a result of cooperation between TripAdvisor and third-party review platforms (e.g., Easytobook.com, Interep, Amoma.com, cheaprooms.com). In this case, solicited reviews are not only displayed on TripAdvisor but also shown on these third-party platforms. At the end of the reviews, there is a disclaimer that states, "Review collected in partnership with (the name of the third-party platform)". In our sample, there are 1,417 reviews collected in this way. Given that review collection in partnership with third-party platforms and hotels can induce different effects, we exclude the 150 hotels whose first solicited review was collected in partnership with third-party platforms and redo our analyses. The results, as reported in Tables A.3 and A.4 in the Online Appendix, show that our findings are not sensitive to the exclusion of these hotels.

Third, it is possible that the implementation of the review solicitation program may be subject to fraudulent practices (e.g., selectively soliciting reviews from guests who have had a positive experience, offering financial incentives) of some hotels. However, such practices should not be common as they are strictly prohibited by TripAdvisor and can result in penalties.[8] To further rule out this possibility, we restrict our analysis to the subsample of chain hotels only as they are unlikely to engage in fraudulent behaviors compared with independent hotels (Mayzlin et al. 2014). The results obtained using this subsample, as shown in Tables A.5 and A.6 in the Online Appendix, are largely consistent with our main results.

5.3.3. Changes in Review Environment. One potential concern with our results is that changes in the review environment (e.g., review volume, average rating) may affect the number of consumers who are eligible to write reviews, consumers' propensity to leave reviews, and the characteristics of the reviews. If hotels' decision to solicit reviews happens to coincide with the changes in these variables, our results might be biased. Following Proserpio and Zervas (2017), we add three variables of the review environment on each platform—log number, average rating, and log average length of previous reviews—as controls and reestimate our cross-platform two-stage DID model. The results, as reported in Tables A.7 and A.8 in the Online Appendix, show that our findings are robust to the inclusion of review environment variables. However, we need to interpret the estimated coefficients with caution since the inclusion of these review environment controls is likely to result in an underestimation of the true effects. The reason for this is that the changes in the review environment may also be caused by the review solicitation and become part of the treatment effects.

5.3.4. Cross-Platform DDD. Thus far, we used only the sample of participating hotels and relied on the DID model for estimation of the treatment effects. As our final robustness test, we use the sample of both participating and nonparticipating hotels and employ the cross-platform two-stage DDD model in Section 4.3 for identification. We report the results in Tables A.9 and A.10 in the Online Appendix. The DDD estimates for the effect of review solicitation on the volume and characteristics of both all reviews and organic reviews are highly consistent with the DID estimates shown in Tables 3 and 4, which provides further support for the validity of our primary identification strategy.

6. Underlying Mechanisms of Negative Spillover: Motivational Crowding-Out

In Section 5.2, we found a negative spillover effect of review solicitation on the volume of organic reviews. In this section, we explore the causes that may lead to such a negative spillover. The negative spillover of review solicitation on the volume of organic reviews can be attributed to two possible explanations. First, reviewers who may have intended to leave an organic review after their stays decide to provide a solicited review after receiving the solicitation emails. We call this the "substitution effect" (i.e., solicited reviews may substitute a set of reviews that reviewers originally plan to contribute voluntarily). Second, review solicitation and the presence of solicited reviews with disclosure may crowd out the intrinsic motivation of organic reviewers such that they decide not to leave a review. We call this the "motivational crowding-out effect." The substitution effect is straightforward and does not reveal anything about the motivational changes of reviewers. Here, we provide two sets of supporting evidence that the motivational crowding-out effect plays an important role in driving this negative spillover.

6.1. Total Number of Reviews Posted by Organic Reviewers Decreased

If the negative spillover can be simply explained by the substitution effect, we should expect an increase or no change in the total number of reviews posted by organic reviewers. If, however, we observe a decrease in the total number of reviews posted by organic reviewers, this can be taken as evidence that the motivational crowding-out effect is driving the negative spillover. To test this, we estimate the impact of review solicitation on all reviews posted by organic reviewers whose first reviews were organic[9] using Equation (1). The results in Table 5 show that the total number of reviews posted by organic reviewers is reduced by 11.7% after review solicitation. This might be a conservative estimate of the motivational crowding-out as it also includes solicited reviews that organic reviewers would not contribute in the absence of solicitation. In view of this evidence, we argue that the motivational crowding-out of organic reviewers is an important driver of the negative spillover of the solicitation on review volume.

6.2. Organic Reviewers Who Never Wrote a Solicited Review Are Influenced More

Next, review solicitation may affect organic reviewers who provided solicited reviews and those who did not differently. Organic reviewers who agreed to write solicited reviews (i.e., impure organic reviewers) may be less resistant to review solicitation than those who refused to do so (i.e., pure organic reviewers). As a result, the intrinsic motivation of pure organic reviewers may be more severely undermined than that of impure organic reviewers. In other words, we expect the motivational crowding-out effect to be stronger for pure organic reviewers. In addition, any decrease in the reviews of pure organic reviewers cannot be attributed to the substitution effect, as they never wrote a solicited review.

Table 5. Effect of Review Solicitation on All Reviews Posted by Organic Reviewers (Two-Stage DID)

	Dependent variable		
	log(Volume) (1)	*Avg_Rating* (2)	*log(Avg_length)* (3)
After × *TripAdvisor*	−0.124***	0.064**	−0.025
	(0.026)	(0.022)	(0.024)
TripAdvisor	Yes	Yes	Yes
After	Yes	Yes	Yes
Hotel fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Observations	23,350	19,433	19,291

Notes. Clustered robust standard errors at the hotel level are in parentheses. The average length includes only English language reviews.

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

To assess the possible heterogeneous effect across the two groups of reviewers, we construct a data set of organic reviewers and run analyses. Specifically, we select organic reviewers who had contributed at least one review by the end of 2010. In these 47,801 organic reviewers, 2,982 of them ended up writing at least a solicited review between 2012 and 2014 and 44,819 did not. We employ a reviewer-level DID method to estimate the heterogeneous effects of the review solicitation program. To reduce the self-selection bias, we apply a coarsened exact matching (CEM) approach with a *k*-to-*k* solution to construct a balanced sample of pure and impure organic reviewers with comparable characteristics. The matching variables we use include the log cumulative number of reviews, log average length per review, log average number of helpful votes per review, average rating per review, and reviewer tenure up to the end of 2010. After matching, we obtain a total of 2,406 pure organic reviewers and 2,406 impure organic reviewers.[10] We adopt the following model:

$$DV_{it} = \beta Post_t \times PureOrganic_i + \alpha_i + \tau_t + \varepsilon_{it}, \qquad (4)$$

where DV_{it} indicates the volume, average rating, and average length of organic reviews written by reviewer i in month t. $Post_t$ is a dummy variable that indicates whether the month t is after the launch of the review solicitation program. $PureOrganic_i$ is a dummy that equals one if the reviewer did not write a solicited review and zero otherwise. We also add reviewer fixed effect and year-month fixed effect. The coefficient β measures the relative changes in the reviews written by pure organic reviewers compared with those written by impure organic reviewers after the introduction of the review solicitation program.

Table 6 presents the estimation results. The coefficient of the interaction term in column 1 is negative and significant, suggesting that pure organic reviewers wrote fewer organic reviews than impure organic reviewers after the launch of the program. As the contributions of most reviewers can be sparse and using a log-linear model could bias the results, we run a Poisson regression on the number of reviews and present the results in column 4. The coefficient is again negative with a larger magnitude. This is consistent with the finding in Khern-am-nuai et al. (2018) that reviewers who are driven solely by intrinsic motivation are demotivated more by external interventions, lending further support to the crowding-out effect.

7. Additional Analyses

7.1. Heterogeneous Treatment Effects

Our analyses thus far treat all the hotels that solicit reviews on TripAdvisor as a whole. However, the

Table 6. Policy Effect on Organic Reviews of Reviewers

	Dependent variable			
	log(*Volume*) (1)	*Avg_Rating* (2)	*log*(*Avg_length*) (3)	*Volume* (4) Poisson
Post × *PureOrganic*	−0.012***	−0.002	0.018	−0.097*
	(0.003)	(0.030)	(0.021)	(0.040)
Reviewer fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Observations	230,976	21,943	21,943	171,936

Notes. Clustered robust standard errors at the hotel level are in parentheses. The average length includes only English language reviews.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

propensity to participate in the review solicitation program and the effects of such solicitation may vary across different hotel types. In Section 3.2.1, we showed that participating and nonparticipating hotels differ substantially in various characteristics. In this section, we explore whether the effects of review solicitation are heterogeneous with respect to these hotel characteristics.

We partition the participating hotels using six different moderators and examine their roles. The six moderators are (1) *Chain*, a dummy variable that indicates whether the hotel is a chain hotel; (2) *Expensive*, a binary variable that equals one if the price of the hotel is above the median price of all participants and zero otherwise[11]; (3) *Large*, a binary variable that equals one if the number of rooms offered by the hotel is above the median size of all participants and zero otherwise; (4) *HighClass*, a binary variable that equals one if the star class of the hotel is above the median and zero otherwise; (5) *Popular*, an indicator of whether the number of reviews that hotels receive before treatment is above the median; and (6) *HighRating*, an indicator of whether the average review rating of hotels before treatment is above the median.

We estimate the cross-platform two-stage DID model by adding an interaction term between the treatment and each moderator and report the regression results regarding the effects on all reviews in Table 7. The positive effect of review solicitation on average review rating is weaker for chain, larger, more expensive, higher-class, more popular, and higher-rated hotels. This might be because these hotels often have a higher baseline rating before review solicitation. Review solicitation, however, does not seem to exert any significant heterogeneous effects on the volume and average length of all reviews. Table 8 displays the results regarding the spillover effect on organic reviews. We do not find any significant heterogeneity in the spillover effect of review solicitation on volume, rating, and length of organic reviews with respect to the six moderators. Combining these results with the comparison in Table 1, we see that hotels that are more likely to participate in the review solicitation program actually benefit less from participating (in terms of the lift in review rating).

7.2. Reviewer-Level Analyses

Individuals may differ substantially in their decision to write solicited reviews. Some users are more willing to do a favor for the hotels by providing their responses, whereas others may be indifferent to or against the solicitation. Upon participation, the solicited reviews written by some reviewers may deviate more from their organic reviews than others. In the following, we perform analyses at the reviewer level to provide insights into these potential heterogeneities.

7.2.1. Who Are More Likely to Write Solicited Reviews?

We first examine what types of reviewers are more likely to provide solicited reviews. To this end, we compiled a sample of 18,520 reviewers who had written at least one review in the last three months before TripAdvisor's launch of the review collection program. Among them, 4,983 reviewers ended up writing solicited reviews, whereas the rest did not. We use both a Cox proportional hazard model and a logit model to examine the relationship between various reviewer characteristics and the decision to provide solicited reviews. We are interested in both self-reported demographic variables including age, gender, and culture,[12] as well as platform-specific behavioral variables including cumulative number of reviews, average review length, average review rating, and cumulative number of helpful votes received right before the policy implementation.

Table 9 presents the results obtained using the Cox proportional hazard model (columns 1 and 2) and the logit model (column 3). The results are consistent between the two models and reveal several interesting findings. First, male reviewers are less likely to write solicited reviews compared with females. This is aligned with the existing research suggesting that females are more empathic and engage more in prosocial and cooperative behaviors to help other individuals and businesses (Eckel and Grossman 2008, Christov-Moore et al. 2014). Second, younger reviewers and privacy-conscious reviewers who did not disclose their age are less likely to write solicited reviews compared with older reviewers. This is also consistent with the work

Table 7. Heterogeneous Effect of Review Solicitation on All Reviews (Two-Stage DID)

	Dependent variable: *log*(*Volume*)					
	Chain (1)	*Expensive* (2)	*Large* (3)	*HighClass* (4)	*Popular* (5)	*HighRating* (6)
After × *TripAdvisor*	0.334***	0.290***	0.334***	0.295***	0.380***	0.329***
	(0.058)	(0.039)	(0.039)	(0.033)	(0.041)	(0.041)
After × *TripAdvisor* × *Moderator*	−0.049	0.006	−0.081	−0.006	−0.129*	−0.092
	(0.067)	(0.058)	(0.058)	(0.070)	(0.058)	(0.059)
	Dependent variable: *Avg_Rating*					
	Chain (1)	*Expensive* (2)	*Large* (3)	*HighClass* (4)	*Popular* (5)	*HighRating* (6)
After × *TripAdvisor*	0.282***	0.202***	0.218***	0.178***	0.230***	0.270***
	(0.052)	(0.041)	(0.037)	(0.029)	(0.045)	(0.038)
After × *TripAdvisor* × *Moderator*	−0.161**	−0.099*	−0.124**	−0.092*	−0.128*	−0.206***
	(0.057)	(0.047)	(0.045)	(0.040)	(0.051)	(0.045)
	Dependent variable: *log*(*Avg_Length*)					
	Chain (1)	*Expensive* (2)	*Large* (3)	*HighClass* (4)	*Popular* (5)	*HighRating* (6)
After × *TripAdvisor*	−0.113*	−0.183***	−0.197***	−0.187***	−0.218***	−0.179***
	(0.052)	(0.040)	(0.040)	(0.031)	(0.046)	(0.037)
After × *TripAdvisor* × *Moderator*	−0.088	0.001	0.017	0.007	0.065	0.021
	(0.059)	(0.050)	(0.050)	(0.049)	(0.053)	(0.050)
TripAdvisor	Yes	Yes	Yes	Yes	Yes	Yes
After	Yes	Yes	Yes	Yes	Yes	Yes
After × *TripAdvisor*	Yes	Yes	Yes	Yes	Yes	Yes
Hotel fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Clustered robust standard errors at the hotel level are in parentheses. The observations for each dependent variable in each subtable are the same as in Table 3.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

revealing a positive impact of age on emotional empathy and prosocial behavior (Sze et al. 2012, Beadle et al. 2015). Third, reviewers from individualistic cultures are more likely to write solicited reviews compared with those from collectivistic cultures. This is in line with the theory that people from individualistic cultures tend to value freedom of expression more. Fourth, more reputable reviewers who write longer reviews and have accumulated more helpful votes are less likely to write solicited reviews compared with less reputable ones. This makes sense as reviewers who are used to contributing high-quality organic reviews may care more about their reputation and hold a negative attitude toward review solicitation. Last, nicer reviewers who tend to give higher ratings in their organic reviews are more likely to write solicited reviews compared with more critical reviewers. This is aligned with our intuition that nicer reviewers are more willing to help hotels by providing solicited reviews with higher ratings. Nevertheless, we need to interpret these results with caution as the revealed relationships may be correlational and could be subject to alternative explanations such as solicitation bias.[13]

7.2.2. Whose Reviews Are Influenced More by Solicitation? Although we know that solicitation positively influences review rating and negatively influences review length overall, it is not clear how these influences may vary among users with different characteristics. In this section, we examine how reviewers' contribution level, age, gender, and culture might moderate the impact of solicitation on review rating and length. For this analysis, we focus on organic reviewers who have posted both organic and solicited hotel reviews on TripAdvisor. We obtained a total of 36,937 organic reviewers who contributed 656,621 reviews. We run a series of regressions with hotel fixed effect and year-month fixed effect.

Columns 1 and 2 of Table 10 present the results for review rating without and with controlling for reviewer fixed effects. The coefficients of $Solicited_{ijt}$ are significantly positive in both models, indicating that the same reviewer tends to give a more positive rating when being asked to provide a review, which is consistent with our findings at the hotel level. From column 1, we also find that reviewers who are older, female, lower level, and from individualistic cultures tend to give

Table 8. Heterogeneous Effect of Review Solicitation on Organic Reviews (Two-Stage DID)

	Dependent variable: *log*(*Volume*)					
	Chain (1)	*Expensive* (2)	*Large* (3)	*HighClass* (4)	*Popular* (5)	*HighRating* (6)
After × *TripAdvisor*	−0.087*	−0.189***	−0.163***	−0.178***	−0.148***	−0.129***
	(0.044)	(0.034)	(0.034)	(0.028)	(0.036)	(0.034)
After × *TripAdvisor* × *Moderator*	−0.102	0.037	−0.013	0.028	−0.000	−0.069
	(0.053)	(0.052)	(0.052)	(0.064)	(0.051)	(0.050)
	Dependent variable: *Avg_Rating*					
	Chain (1)	*Expensive* (2)	*Large* (3)	*HighClass* (4)	*Popular* (5)	*HighRating* (6)
After × *TripAdvisor*	0.076	0.079	0.104**	0.053	0.089	0.115**
	(0.054)	(0.042)	(0.039)	(0.030)	(0.049)	(0.040)
After × *TripAdvisor* × *Moderator*	−0.044	−0.073	−0.119*	−0.045	−0.070	−0.126**
	(0.059)	(0.048)	(0.047)	(0.042)	(0.054)	(0.047)
	Dependent variable: *log*(*Avg_Length*)					
	Chain (1)	*Expensive* (2)	*Large* (3)	*HighClass* (4)	*Popular* (5)	*HighRating* (6)
After × *TripAdvisor*	0.039	0.041	0.024	0.013	0.021	0.030
	(0.052)	(0.038)	(0.039)	(0.030)	(0.044)	(0.035)
After × *TripAdvisor* × *Moderator*	−0.043	−0.060	−0.040	−0.025	−0.024	−0.023
	(0.058)	(0.047)	(0.047)	(0.044)	(0.051)	(0.047)
TripAdvisor	Yes	Yes	Yes	Yes	Yes	Yes
After	Yes	Yes	Yes	Yes	Yes	Yes
After × *TripAdvisor*	Yes	Yes	Yes	Yes	Yes	Yes
Hotel fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Clustered robust standard errors at the hotel level are in parentheses. The observations for each dependent variable in each subtable are the same as in Table 4.

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

higher ratings. These results are consistent with those reported in previous studies. The coefficients of the interaction terms, which represent our coefficients of interest, reveal that the positive impact of solicitation on review rating is stronger for female and older reviewers.

Columns 3 and 4 show the results for review length. The coefficients of $Solicited_{ijt}$ are significantly negative in both models, confirming our expectation that the same reviewer tends to write a shorter review when being solicited for a review. From column 3, we observe that higher-level reviewers, females, younger reviewers, and those from individualistic cultures tend to write longer reviews. The coefficients of interaction terms indicate that the negative impact of solicitation on review length is stronger for lower-level contributors, female reviewers, and reviewers from individualistic cultures.

Taken together, we provide empirical evidence that the review ratings of female and older reviewers are influenced more by solicitation, and the review length of female reviewers, lower-level reviewers, and reviewers from individualistic cultures are influenced more by solicitation. Combining this set of results with the findings in Section 7.2.1, it appears that individuals who are more likely to write solicitation reviews are also the ones whose review characteristics are influenced more by solicitation. A plausible explanation could be that reviewers who have a higher propensity to provide solicited reviews would do so even when they are less intrinsically motivated, resulting in solicited reviews that are substantially different from their organic reviews.

7.3. Shift in Review Topics

Most of the previous research on review solicitation has focused on the effects on review characteristics such as length, valence, helpfulness, readability, and emotion. Very little research has examined the actual content or topics discussed in the review, which is crucial as it can directly affect the perceived informativeness and value of the review. Email solicitation, an approach to nudge consumers to provide reviews, may also result in a topic shift in the review content.

In this section, we are interested in identifying the potential differences in the topic distributions between solicited and organic reviews. We use the STM model proposed by Roberts et al. (2013, 2014), which is a recently developed tool that provides a flexible way to incorporate document metadata into the topic modeling

Table 9. Who Are More Likely to Write Solicited Reviews

	Cox proportional hazard model		Logit model
	Parameter estimate (p value) (1)	Hazard ratio (95% CI for HR) (2)	Parameter estimate (p value) (3)
Gender: no disclosure	−0.114	0.908	−0.097
	(0.065)	[0.816, 1.010]	(0.0543)
Gender: man	−0.096*	0.914*	−0.090*
	(0.043)	[0.853, 0.980]	(0.036)
Age: no disclosure	−0.623***	0.598***	−0.515***
	(0.084)	[0.524, 0.682]	(0.067)
Age: 18–34	−1.500***	0.270***	−1.309***
	(0.102)	[0.227, 0.321]	(0.089)
Age: 35–49	−0.911***	0.461***	−0.775***
	(0.075)	[0.409, 0.518]	(0.060)
Age: 50–64	−0.203**	0.843**	−0.171**
	(0.073)	[0.755, 0.941]	(0.057)
Individualism	0.003*	1.003*	0.003*
	(0.001)	[1.000, 1.005]	(0.001)
log(Cumu_Num)	−0.045	0.966	−0.034
	(0.044)	[0.898, 1.040]	(0.037)
log(Avg_Length)	−0.129***	0.891***	−0.115***
	(0.028)	[0.852, 0.931]	(0.023)
Avg_Rating	0.122***	1.106***	0.101***
	(0.026)	[1.059, 1.156]	(0.022)
log(Cumu_Votes)	−0.088*	0.923*	−0.080*
	(0.038)	[0.867, 0.982]	(0.032)
N	17,376		17,376
Log likelihood	−45,254.95		−9,862.996

Note. Robust standard errors are in parentheses.
***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

process. Like other topic models such as the latent Dirichlet allocation (LDA) (Blei et al. 2003), in the STM, each document is represented as a mixture over a set of topics and each topic is a distribution over words. However, different from the vanilla LDA model, STM allows topics to be correlated with each other and the topical prevalence (i.e., the frequency with which a topic is discussed) and topical content (i.e., the words used to discuss a topic) to vary with document-level covariates. By including the solicitation status of a review into the covariates on topical prevalence, STM allows us to gain valuable insights into whether solicitation causes any systematic changes in how often particular topics are discussed in a review.

The data in this section contain 247,414 English reviews from 554 participating hotels on TripAdvisor from January 2012 to January 2017, including a total of 150,873 organic reviews and 96,541 solicited reviews. We adopt a set of standard preprocessing procedures such as stemming and removing punctuation and stop words. The remaining words are used to construct a term-document matrix where each cell represents the number of times a term indicated by the column appears in a document indicated by the row.

In our context, each document is a review. We allow the topical prevalence of a review to vary with solicitation status, hotel, year-month, rating, and length so that we can accurately measure how solicitation status affects the frequency with which a specific topic is discussed while controlling for other factors. Next, we need to choose the optimal number of topics. We fit the STM model with the number of topics ranging from 5 to 50 with a step size of 5 and determine the optimal value based on two criteria: semantic coherence and exclusivity (Mimno et al. 2011, Roberts et al. 2014). Semantic coherence measures the degree to which high-probability words in a given topic co-occur, whereas exclusivity measures the extent to which the top words in one topic do not appear as top words in other topics. A topic with high coherence and exclusivity is more likely to be semantically interpretable (Roberts et al. 2014). We plot the average semantic coherence and exclusivity of the obtained models in Figure B.1 in the Online Appendix. We set the number of topics to 25 because it achieves reasonably high scores in both semantic coherence and exclusivity.

Topic Summary. We fit the STM model with 25 topics and present the results in Table B.1 in the Online Appendix. Column 2 shows the top ten words of each topic with the highest frequency exclusivity (FREX). By looking at these top words, we can clearly see that the topics are highly interpretable and distinct from one another. In column 1, we assign a label for each topic based on

Table 10. Whose Reviews Are Influenced More by Solicitation

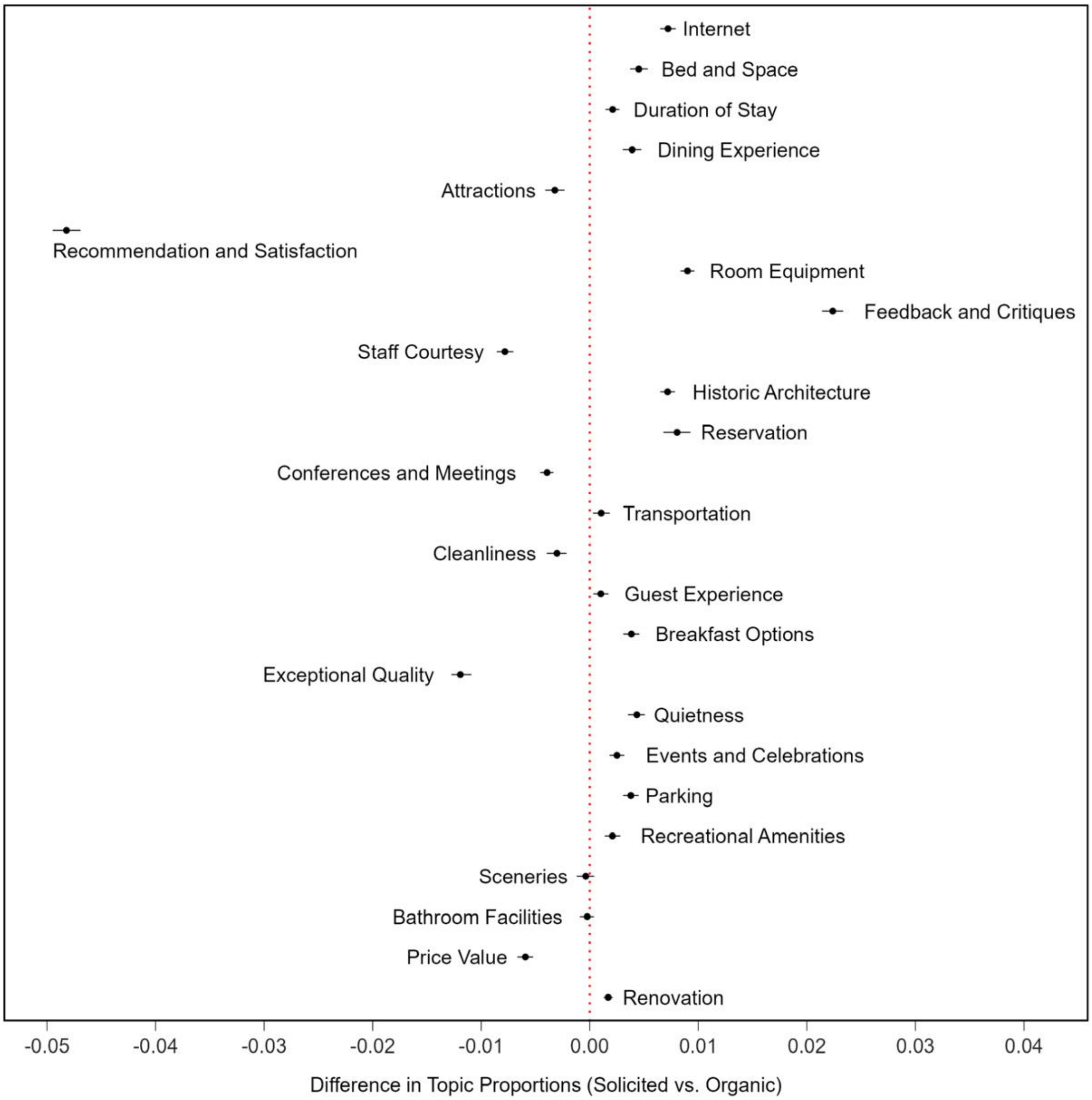
	Dependent variable: *Rating*		Dependent variable: *log*(*Length*)	
	(1)	(2)	(3)	(4)
Solicited	0.207***	0.212***	−0.295***	−0.320***
	(0.043)	(0.041)	(0.025)	(0.024)
Reviewer_Level	−0.020***		0.023***	
	(0.002)		(0.001)	
Gender: no disclosure	−0.022***		−0.100***	
	(0.005)		(0.004)	
Gender: man	−0.066***		−0.089***	
	(0.003)		(0.003)	
Age: no disclosure	−0.079***		0.012**	
	(0.006)		(0.004)	
Age: 18–34	−0.134***		0.139***	
	(0.009)		(0.007)	
Age: 35–49	−0.142***		0.095***	
	(0.006)		(0.004)	
Age: 50–64	−0.079***		0.035***	
	(0.005)		(0.004)	
Individualism	0.001***		0.004***	
	(0.000)		(0.000)	
Solicited × *Reviewer_Level*	0.001	−0.007*	0.010***	0.015***
	(0.003)	(0.003)	(0.002)	(0.002)
Solicited × *Gender: no disclosure*	0.001	−0.038*	0.024**	0.015
	(0.015)	(0.015)	(0.009)	(0.009)
Solicited × *Gender: man*	−0.022	−0.058***	0.026***	0.012
	(0.012)	(0.012)	(0.007)	(0.007)
Solicited × *Age: no disclosure*	−0.031	−0.035	0.013	0.025*
	(0.020)	(0.020)	(0.012)	(0.012)
Solicited × *Age: 18–34*	−0.145***	−0.159***	−0.007	−0.021
	(0.032)	(0.031)	(0.019)	(0.019)
Solicited × *Age: 35–49*	−0.063**	−0.056**	−0.022	−0.031*
	(0.020)	(0.020)	(0.012)	(0.012)
Solicited × *Age: 50–64*	−0.023	−0.030	−0.016	−0.022*
	(0.018)	(0.018)	(0.011)	(0.011)
Solicited × *Individualism*	0.000	0.000	−0.002***	−0.002***
	(0.000)	(0.000)	(0.000)	(0.000)
Reviewer fixed effects	/	Yes	/	Yes
Hotel fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Observations	568,045	567,795	568,045	567,795
R^2	0.273	0.442	0.262	0.589

Notes. Clustered robust standard errors at the hotel level are in parentheses. The average length includes only English language reviews.
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$.

the top words that appear in the topic. Column 3 presents the estimated proportions of each topic in the reviews.

Topic Shift. A core advantage of the STM model is the ability to estimate and visualize the effect of interested covariates on topic distributions. We first estimate the total effect of solicitation with hotel and year-month fixed effects.[14] Figure 4 plots the changes in topic proportions from organic reviews to solicited reviews. We see that the proportion of the topic "Recommendation and Satisfaction" in solicited reviews is markedly higher than that in organic reviews. That is, consumers are much more likely to discuss "Recommendation and Satisfaction" when being solicited for a review compared with when they contribute a review organically. The other six topics that also attract more attention in solicited reviews are "Exceptional Quality," "Staff Courtesy," "Price Value," "Conferences and Meetings," "Attractions," and "Cleanliness." On the other hand, there are 16 topics that are discussed more in organic reviews, which are "Feedback and Critiques," "Room Equipment," "Reservation," "Internet," "Historic Architecture," "Bed and Space," "Quietness," "Dining Experience," "Breakfast Options," "Parking," "Events and Celebrations," "Duration of Stay," "Recreational Amenities," "Renovation," "Transportation," and "Guest Experience." These results indicate that reviewers tend to discuss their attitudes toward hotels when being solicited to contribute a review. However, specific details of hotels such as services, facilities, design, and food, which often require consumers to exert more effort to recall and articulate their stay experience, are mentioned less. To explore

Figure 4. (Color online) Difference in the Topic Proportions (Organic − Solicited)



whether the observed solicitation impact on review topics could simply be attributed to changes in review rating and length, we re-estimate the effect while controlling for rating and length. Figure B.2 in the Online Appendix plots the changes in topic proportions from organic reviews to solicited reviews under this estimation. We can see that even after excluding the mediation effect of rating and length, there remains a significant direct influence of solicitation on the shift in review content from specific and concrete topics to general and abstract topics. These findings are likely caused by reviewers' lower level of intrinsic motivation when composing solicited reviews.[15] The shift in review content from more specific topics to more general comments would certainly reduce the informativeness and value of reviews, which can be harmful for the platform, hotels, and consumers.

To further validate the results from STM that the content of organic reviews is more specific and concrete, we use the GPT, a large language model (LLM) developed by OpenAI, to annotate hotel reviews.[16] Recent work has demonstrated that LLMs consistently outperform crowd workers for text annotation (Gilardi et al. 2023, Törnberg 2023). To evaluate the total effect of the solicitation on review specificity, we performed exact matching between organic and solicited reviews on two criteria: hotel and year-month. This process yielded a total of 52,634 pairs of organic and solicited reviews. Of these, we randomly selected 10,000 pairs for GPT validation. We then used the gpt-3.5-turbo version of the GPT API to conduct the annotation task for 10,000 iterations. In each iteration, we presented a matched pair of organic and solicited reviews and posed the following question: "Which review focuses more on specific features and provides more detailed information of the hotel?" Our results show that the GPT model recognized organic reviews as being more specific and detailed in 78.97% of trials. To validate whether solicitation exerts a direct impact beyond its potential indirect effect mediated through rating and length, we strengthened our matching criteria by requiring that the paired reviews not only originate from the same hotel and year-month, but also share identical ratings and similar lengths. This stringent matching gave us a total of 25,119 pairs of organic and solicited reviews.[17] This time, the GPT model identified organic reviews as being more specific and detailed in 66.49% of trials. These results corroborate the STM findings that solicited reviews tend to be less specific and concrete.

8. Discussion and Conclusion

As consumers increasingly rely on user-generated reviews to make their purchase decisions, review platforms and businesses are adopting a wide range of strategies to stimulate online reviews. In this paper, we study the effect of hotels' participation in a review solicitation program endorsed by TripAdvisor on subsequent review generation. By applying a rigorous cross-platform two-stage DID (and DDD) model on reviews posted on TripAdvisor and Expedia, we confirm that this program is indeed effective in boosting the number of total reviews. However, this comes at the cost of reduced review quality in that the reviews are much shorter and positively inflated. Given that consumers perceive longer and more negative reviews to be more valuable (Schindler and Bickart 2012, Yin et al. 2016, Eslami et al. 2018), this suggests a reduction in the general value of reviews. In addition, we identify that review solicitation creates a notable negative spillover on the volume of organic reviews and present evidence that the motivational crowd-out of organic reviewers is an important driver of this spillover. Additional analyses show that the overall effect of review solicitation is heterogeneous across hotels of different types. In addition, consumer responses to review solicitation in terms of the propensity to write solicited reviews and the deviation in review characteristics from organic reviews vary considerably with their demographic and behavioral characteristics. Last, by mining the textual content of reviews using an advanced STM model, we discover another pitfall of review solicitation that it causes a significant shift in review content from specific and concrete topics to general and abstract topics.

Our findings have important implications for business managers who are considering participating in platform-initiated review solicitation programs. We show that participation is likely to benefit firms in review volume and valence, but the effect on review valence is heterogeneous across businesses of different types. Specifically, popular businesses with high ratings do not enjoy as much lift in review valance as less popular businesses with low ratings. Therefore, there is less need for these businesses to adopt the practice of review solicitation. Despite the apparent benefits, businesses should be mindful of the unintended consequences that may arise. For example, solicitation may hurt reviewers' intrinsic motivation and lead to a reduction in the length of reviews. In addition, the information diagnosticity of reviews becomes lower as reviewers move away from a thorough reflection of the details of their stay experience. If businesses decide to solicit reviews, they may want to frame the solicitation messages to encourage longer and more informative reviews, such as by using social norms (Burtch et al. 2018) or giving consumers detailed guidance on review writing. In addition, solicited reviews may crowd out the provision of organic reviews—which are perceived as more trustworthy and credible by consumers—and hurt consumers' brand attitude and evaluation. Therefore, we encourage businesses to carefully weigh the benefits against potential harms associated with participating in review solicitation before making their final decision.

Our research also provides useful insights for owners of review platforms. Given the nonincentivized nature and explicit disclosure of solicitation information, the platform-initiated review solicitation program may provide a convenient and legitimate way for platform owners to solve the issue of under-provision of consumer reviews. Nonetheless, as revealed in this paper, the influx of positively biased and less informative solicited reviews, along with their negative spillover on the generation of organic reviews, emphasizes the necessity for platforms to approach this solicitation strategy with caution. We urge platform owners to deliberate carefully and undertake a thorough evaluation of the potential benefits and pitfalls before choosing to adopt this approach. When making the assessment, platforms should also consider the demographic and behavioral characteristics of their target users and their likely reactions to such solicitation. If platforms opt to implement the review solicitation program, they should consider devising effective strategies to mitigate any potential adverse impacts. Given the risk that positively inflated solicited reviews could compromise the fairness of recommendation and ranking systems, and possibly mislead consumers in their choice of products and services, review platforms could develop proprietary algorithms to correct this bias and provide more relevant recommendations to consumers. Moreover, the presence of brief and uninformative solicited reviews might undermine the long-term value and competitiveness of review platforms. One potential solution is to offer consumers the option to filter reviews by whether they are organic or solicited. This would give consumers the discretionary power to decide the extent to which they want to rely on each type of review. To alleviate consumer skepticism, platforms should craft the solicitation disclosure in a way that emphasizes the nonincentivized nature of the reviews. We also recommend that platforms acknowledge the potential bias in solicited reviews and emphasize that organic reviews of the participating businesses are not subject to this bias. Finally, to combat the motivational crowd-out of organic reviewers, platforms can advocate that platform-initiated review solicitation, with the guidance of trusted partners, can reduce firms' likelihood of soliciting more biased incentivized reviews, and thereby help promote a more sustainable community.

Although our focus in this paper is a travel and accommodation platform, we believe our results can be generalized to other types of review platforms, such as e-commerce sites, app stores, and sharing economy platforms. This is because of the widespread nature of the motivations driving users' provision of online reviews. Although the degree of these motivations may vary with the products and services being reviewed, we expect that the fundamental mechanisms underpinning the findings in this paper will persist across diverse platforms. Nevertheless, we acknowledge that the platform-initiated review solicitation program we study only represents one specific form of review solicitation. In practical settings, the nature of review solicitation can vary considerably across aspects such as the offering of financial incentives, the inclusion of a solicitation disclosure, and the senders of solicitation messages, which could be either platforms or businesses. The generalizability of our findings to other forms of review solicitation may need further validation.

Our study is subject to several limitations, which also provide opportunities for future work. First, we estimate the ATT, which captures the effects of review solicitation conditional on the hotels that participate in the program. We need to interpret this result with caution as it can differ from the ATE due to self-selection bias. In addition, the design and framing of the solicitation emails might affect both recipients' intention to write a review and the valence and quality of the solicited reviews. Future research could conduct experiments to examine how different framings of solicitation messages might lead to possible heterogeneous effects. Second, the way in which the solicitation disclosure is formulated could also affect users' motivation to contribute reviews. Future work could conduct experiments to explore how the formulation of disclosure would affect the results. Third, although we have provided empirical evidence regarding the generation of solicited reviews, the spillover to organic reviews, and the possible underlying mechanisms, direct measures of consumers' motivations and attitudes toward hotels and review platforms are lacking. Qualitative methods such as interviews and surveys could be used to explicate consumers' intentions and perceptions and enrich our findings. Last, our study does not address the impact of review solicitation on sales, which is an important variable of interest from hotels' perspective. Although review solicitation substantially increases the number of reviews a hotel receives, consumers may be skeptical about the review quality and reviewers' impartiality and may thus be hesitant to make purchase decisions. Future efforts can be devoted to assessing the effectiveness of review solicitation on product sales.

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Endnotes

[1] See https://www.amazon.com/gp/vine/help (accessed March 2023).

[2] See https://www.tripadvisor.com/ReviewExpress (accessed March 2023).

[3] See https://www.yelp-support.com/article/Don-t-Ask-for-Reviews (accessed March 2023).

[4] This information is provided by TripAdvisor. See https://tripadvisor.mediaroom.com/US-about-us (accessed March 2023).

[5] Some hotels in our sample may not have the full observations of 12 months before and after treatment on the two platforms. In Section 5.3.2, we conduct a robustness check by excluding these hotels.

[6] To further validate our choice of Expedia as a control group, we plot the number of reviews for participating hotels on TripAdvisor and Expedia in the 12 months both before and after hotels' participation in TripAdvisor's review solicitation program in Figure A.1 in the Online Appendix. It is apparent that the numbers of hotel reviews on the two platforms followed a parallel trend before participation. After participation, we observe that the number of Expedia reviews still followed the original trend, lending support to the use of Expedia as a control group.

[7] We calculate the effect size using the formula $\exp(0.295) - 1 = 34.3\%$. The same calculation applies to all subsequent effect size estimates for review volume and length.

[8] See https://www.tripadvisor.com/TripAdvisorInsights/w594 (accessed March 2023).

[9] Our findings are robust to the use of two alternative operationalizations of organic reviewers: (1) reviewers who were active before the policy change and (2) reviewers whose majority type of reviews is organic. We present the results in Tables A.11 and A.12 in the Online Appendix.

[10] We conduct a balance check to confirm that there is no significant difference between the two groups after matching.

[11] We define the price of each hotel using the midpoint of its price range displayed on TripAdvisor.

[12] Here, we focus on the cultural dimension of individualism versus collectivism, which has garnered substantial attention in the context of online content generation (Fang et al. 2013, Kayes et al. 2015). We extracted reviewers' self-reported country of residence and measured their level of individualism using Hofstede's individualism

score (Hofstede 2001). We excluded 1,144 reviewers due to missing values in their individualism metric, resulting in a final count of 17,376 reviewers.

[13] There might be variation in the likelihood of different types of reviewers receiving solicitation emails from hotels, potentially leading to disparities in their contribution of solicited reviews.

[14] Considering that rating and length are post-treatment variables that can be influenced by solicitation, they are not included in this estimation.

[15] In Section B.2 of the Online Appendix, we rule out an alternative explanation that solicited reviews tend to be more abstract because they are authored after a greater delay.

[16] We thank the associate editor for making this suggestion.

[17] The average length discrepancy between organic and solicited reviews in matched pairs is less than 2%.

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